Meghna Sengupta

PhD Candidate in Cryptography

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Education

2023–2026 (expected) **PhD, School of Informatics**, *University of Edinburgh*, Edinburgh, UK
Thesis: *Privacy with Accountability: Verifiable and Modular Cryptographic Systems.*
Supervisors: Prof. Markulf Kohlweiss and Prof. Anna Lysyanskaya.

2019–2022 **M.S. in Computer Science**, *Boston University*, Boston, MA, USA

2014–2019 **Dual Degree (B.Tech. and M.Tech.) in Computer Science and Engineering**, *Indian Institute of Technology Kharagpur*, Kharagpur, India, CGPA: 9.6/10

Research Interests

Theoretical cryptography; Privacy-preserving cryptographic systems; Anonymous credentials; Zero-knowledge proofs; Verifiable computation; Provable security.

Publications and Manuscripts

[1] Jan Bobolz, Jesus Diaz Vico, Markulf Kohlweiss, and **Meghna Sengupta**. *Pluggable Anonymous Credentials*. Manuscript in preparation. [draft]

[2] Scott Griffy, Markulf Kohlweiss, Anna Lysyanskaya, and **Meghna Sengupta**. *Succinctly Verifiable Computation over Additively-Homomorphically Encrypted Data: Making Privacy-Preserving Blueprints Practical*. **PKC 2026**, pp. 235–269. [paper]

[3] **Meghna Sengupta**. *Beyond Confidentiality: Framing-Resistant Secure Vault Schemes*. **INDOCRYPT 2025**, pp. 442–460. [paper]

[4] Rishiraj Bhattacharyya, Avradip Mandal, and **Meghna Sengupta**. *Secure Vault Scheme in the Cloud Operating Model*. **INDOCRYPT 2024**, pp. 281–303. [paper]

[5] **Meghna Sengupta**. *Mining out of Delphi: Bitcoin Proofs of work without Random Oracles*. [MS Thesis]

[6] **Meghna Sengupta**, Swapan Maiti, and Dipanwita Roy Chowdhury. *Generating Nonlinear Codes for Multi-bit Symbol Error Correction Using Cellular Automata*. *Physica D: Nonlinear Phenomena*, 415:132758, 2021. [paper]

[7] **Meghna Sengupta**, Swapan Maiti, and Dipanwita Roy Chowdhury. *Nonlinear Error Correcting Codes Using Cellular Automata*. Exploratory paper, **AUTOMATA 2018**, Ghent University, Belgium.

Research Experience

2022–2023 **Research Project — CellTree**, *IIT Bombay*, Mumbai, India, Prof. Manoj Prabhakaran
Worked on extending and developing CellTree, a framework for programmable distributed data repositories with evolving, independently governed cells.

2019–2022 **M.S. Research Project**, *Boston University*, Boston, MA, USA, Prof. Leonid Reyzin
Studied standard-model replacements for random-oracle functionality in Bitcoin proof-of-work, including moderately hard correlation-intractable functions and sequential puzzles.

2020 **Research Intern, Crypto Research Group**, *Fujitsu Research of America*, USA
Investigated meaningful notions of memory-hard functions for blockchain applications and analysed existing candidates under the resulting model.

May–Jul 2018 **DAAD Research Fellow / Summer Research Intern**, *Ruhr University Bochum*, Bochum, Germany, Prof. Gregor Leander
Analysed the algebraic properties of a 128-bit block cipher and derived bounds on the algebraic degree of its round function.

May–Jul 2017 **Summer Research Intern**, *KU Leuven (COSIC)*, Leuven, Belgium, Prof. Ingrid Verbauwhede
Optimised software components of a Ring-LWE encryption implementation, including number-theoretic transforms and modular reduction.

2018–2019 **M.Tech. Project**, *Indian Institute of Technology Kharagpur*, Kharagpur, India, Prof. Abhijit Das
Implemented and optimised batch verification of elliptic-curve digital signatures using SIMD instructions on Koblitz curves.

2017–2018 **B.Tech. Project**, *Indian Institute of Technology Kharagpur*, Kharagpur, India, Prof. Abhijit Das
Implemented and optimised ECDSA and supporting finite-field arithmetic using SIMD instructions.

Jun–Jul 2016 **Summer Research Intern**, *University of Washington*, Seattle, WA, USA, Prof. Arvind Krishnamurthy
Studied resource allocation for network switches using constraint and SAT solvers.

Selected Talks and Academic Activities

2026 **Participant**, *Dagstuhl Seminar 26171: Privacy-Preserving Authentication*, Schloss Dagstuhl, Germany
Invitation-based seminar on anonymous credentials, blind signatures, zero-knowledge proofs, revocation, and privacy-preserving authentication.

2025 **Conference Presentation**, *INDOCRYPT 2025*
Presented *Beyond Confidentiality: Framing-Resistant Secure Vault Schemes*.

2024 **Conference Presentation**, *INDOCRYPT 2024*, Chennai, India
Presented *Secure Vault Scheme in the Cloud Operating Model*.

Teaching Experience

2023, 2024 **Teaching Assistant — Computer Security**, *University of Edinburgh*, Edinburgh, UK
Taught and supported coursework for approximately 70 students.

2020, 2021 **Teaching Assistant — Cryptography and Network Security**, *Boston University*, Boston, MA, USA
Supported approximately 60 students; helped prepare and grade assignments.

2019 **Teaching Assistant — Discrete Mathematics**, *IIT Kharagpur*, Kharagpur, India
Regularly conducted tutorials for approximately 100 students.

2018 **Teaching Assistant — Switching Circuits and Logic Design**, *IIT Kharagpur*, Kharagpur, India
Taught laboratory sessions for approximately 100 students.

Selected Awards and Honours

2019 Graduated among the **top 1% of students** across all branches at IIT Kharagpur, and with the **highest CGPA among female students** in the graduating cohort.

2018 **DAAD Fellowship** for a research internship at Ruhr University Bochum.

2017 **Sikharini Nag Memorial Award**, awarded for the highest CGPA among female students across engineering branches at IIT Kharagpur at the end of the sixth semester.

Service and Interests

2022–2023 **Animal Welfare Committee, IIT Bombay** — active volunteer in animal welfare activities; animal welfare remains a strong personal interest.

2014-2016 **National Service Scheme, IIT Kharagpur** — group leader for a team of 18 volunteers; organised clothing donations and regularly taught children in underserved communities.

Other — 5-year certified course in Fine Arts from Rabindra Charukala Parishad; received multiple gold medals. Avid reader; also interested in travel and the arts.

Technical Skills

Programming Python, C, C++; familiarity with Assembly, Verilog, VHDL, and VBA.

References

Prof. Markulf Kohlweiss Professor, School of Informatics, University of Edinburgh markulf.kohlweiss@ed.ac.uk

Prof. Leonid Reyzin Professor, Department of Computer Science, Boston University reyzin@bu.edu